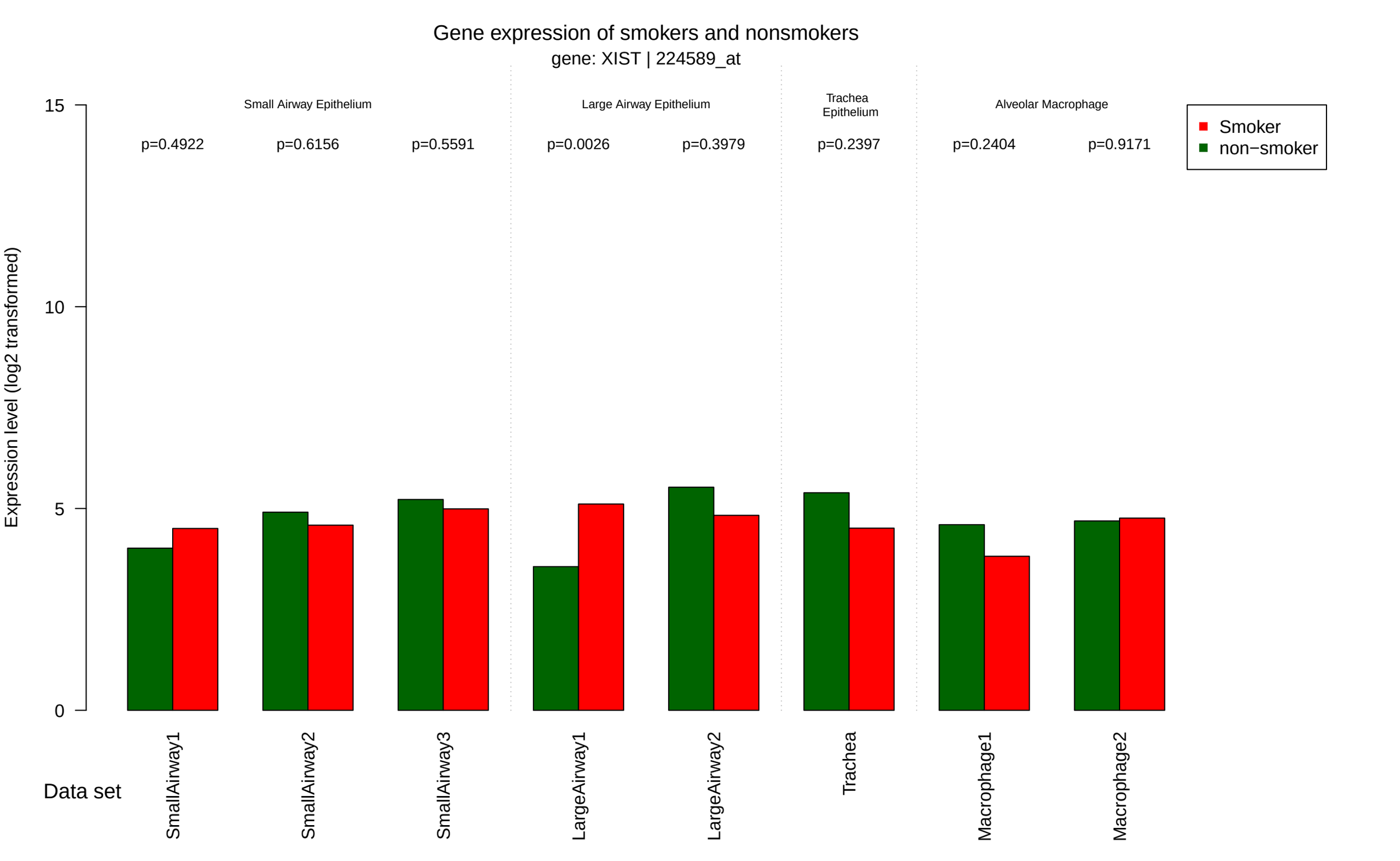
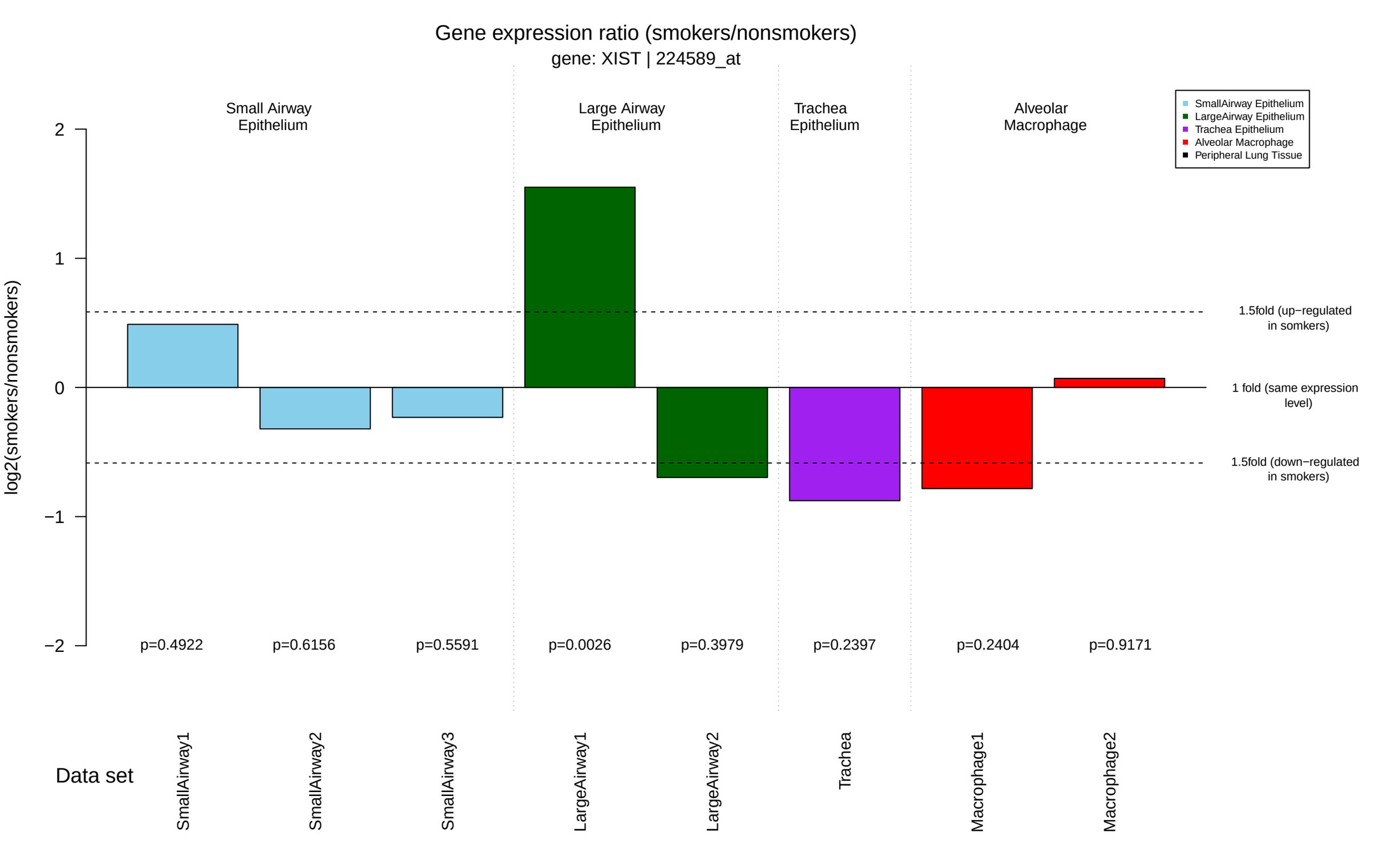
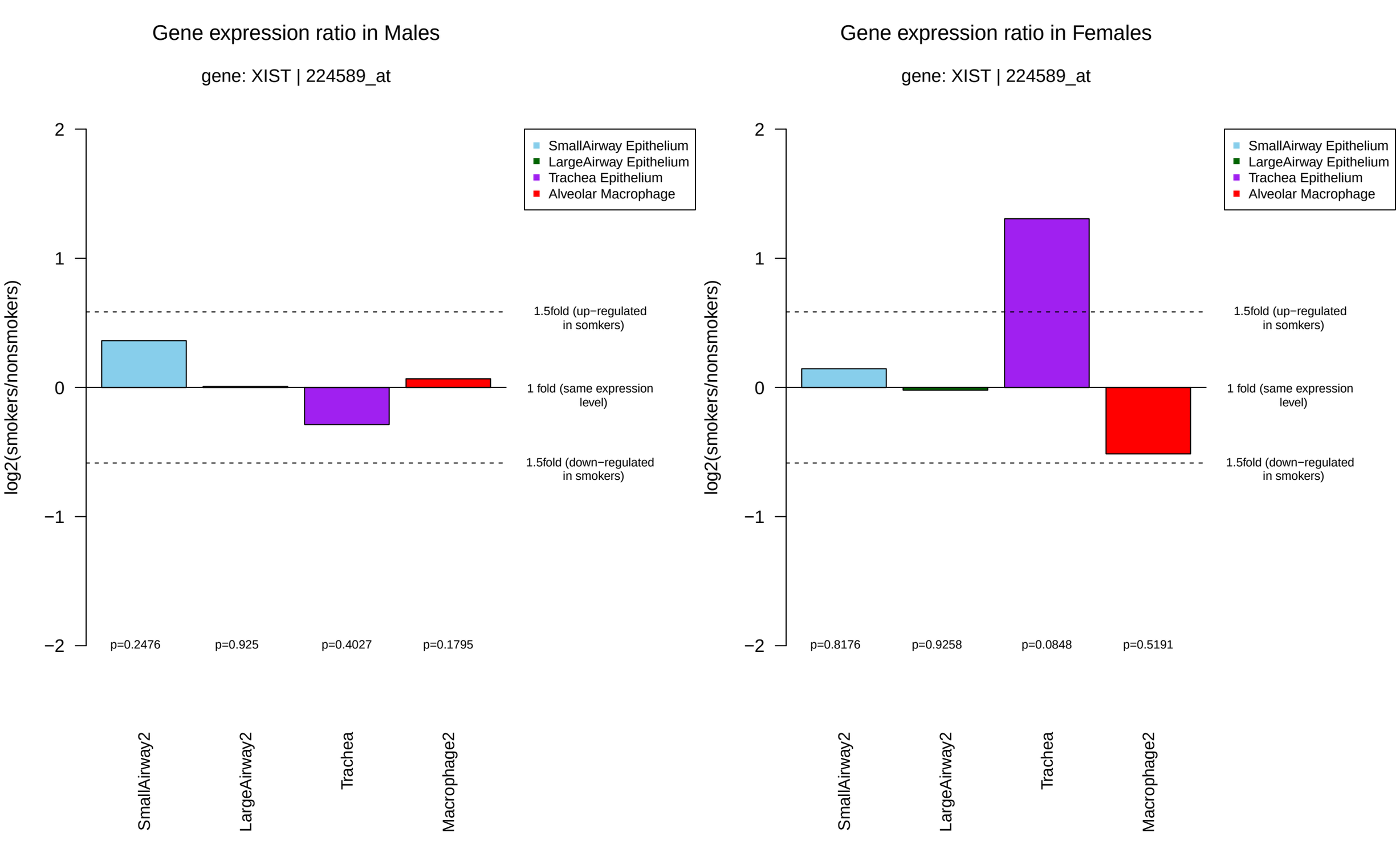
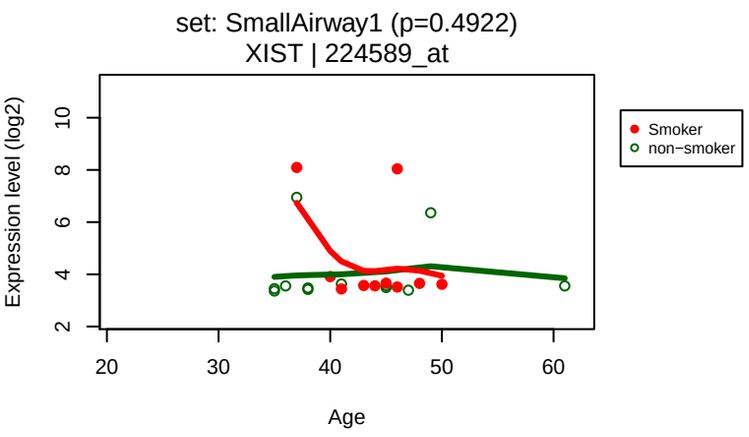
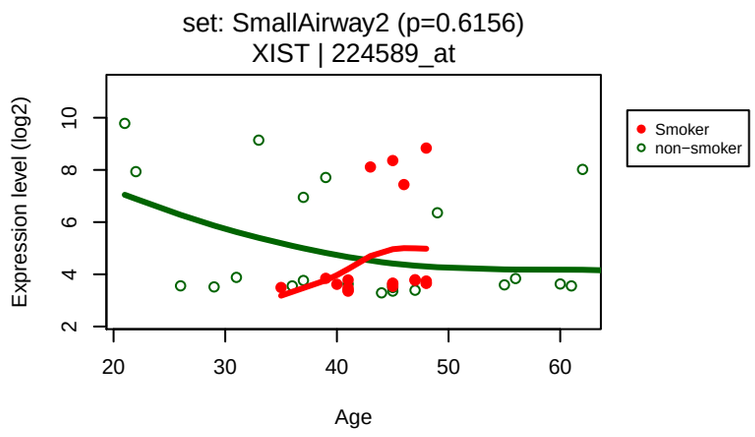
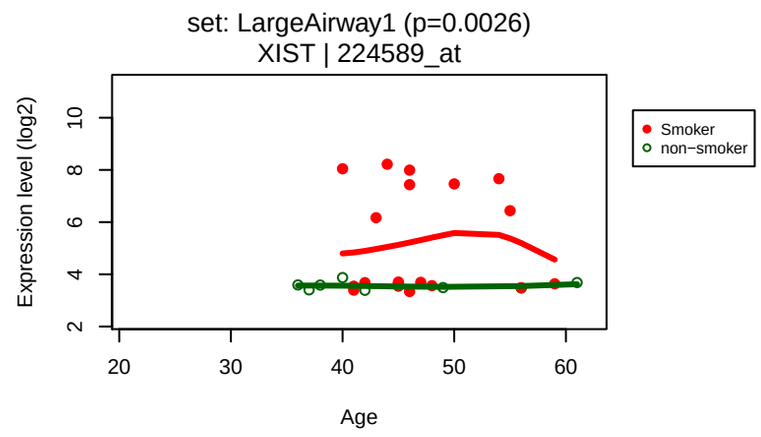

Gene expression of smokers and nonsmokers
gene: XIST | 224589_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4922
p=0.6156
p=0.5591
p=0.0026
p=0.3979
p=0.2397
p=0.2404
p=0.9171
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

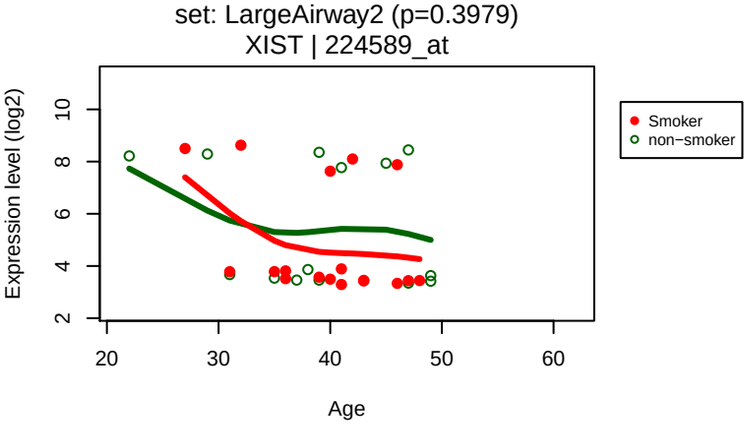

Gene expression ratio (smokers/nonsmokers)
gene: XIST | 224589_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4922
p=0.6156
p=0.5591
p=0.0026
p=0.3979
p=0.2397
p=0.2404
p=0.9171
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

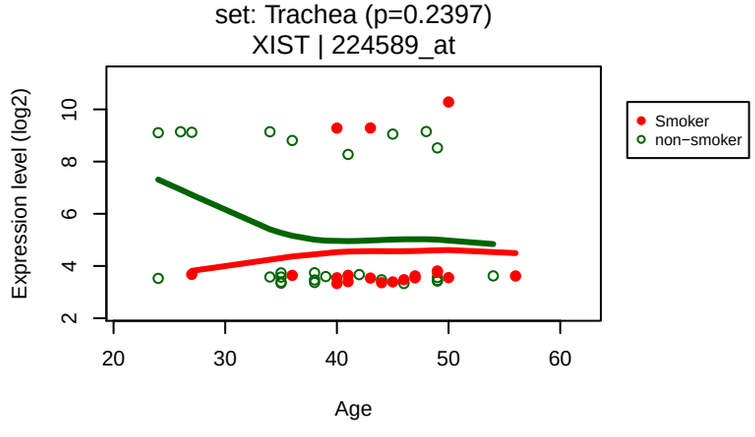

Gene expression ratio in Males
gene: XIST | 224589_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2476
p=0.925
p=0.4027
p=0.1795
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: XIST | 224589_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8176
p=0.9258
p=0.0848
p=0.5191
SmallAirway2
LargeAirway2
Trachea
Macrophage2

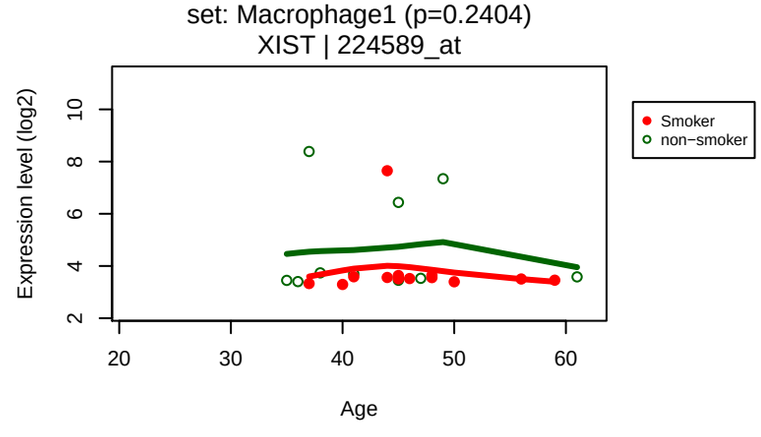

set: SmallAirway1 (p=0.4922)
XIST | 224589_at
Expression level (log2)
Age
Smoker
non−smoker

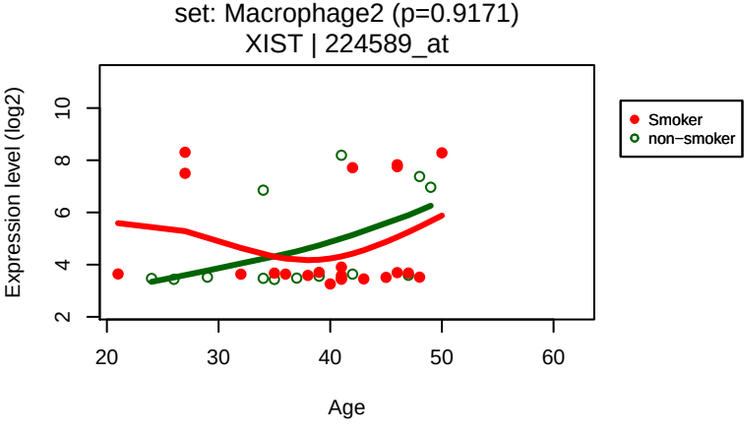

set: SmallAirway2 (p=0.6156)
XIST | 224589_at
Expression level (log2)
Age
Smoker
non−smoker

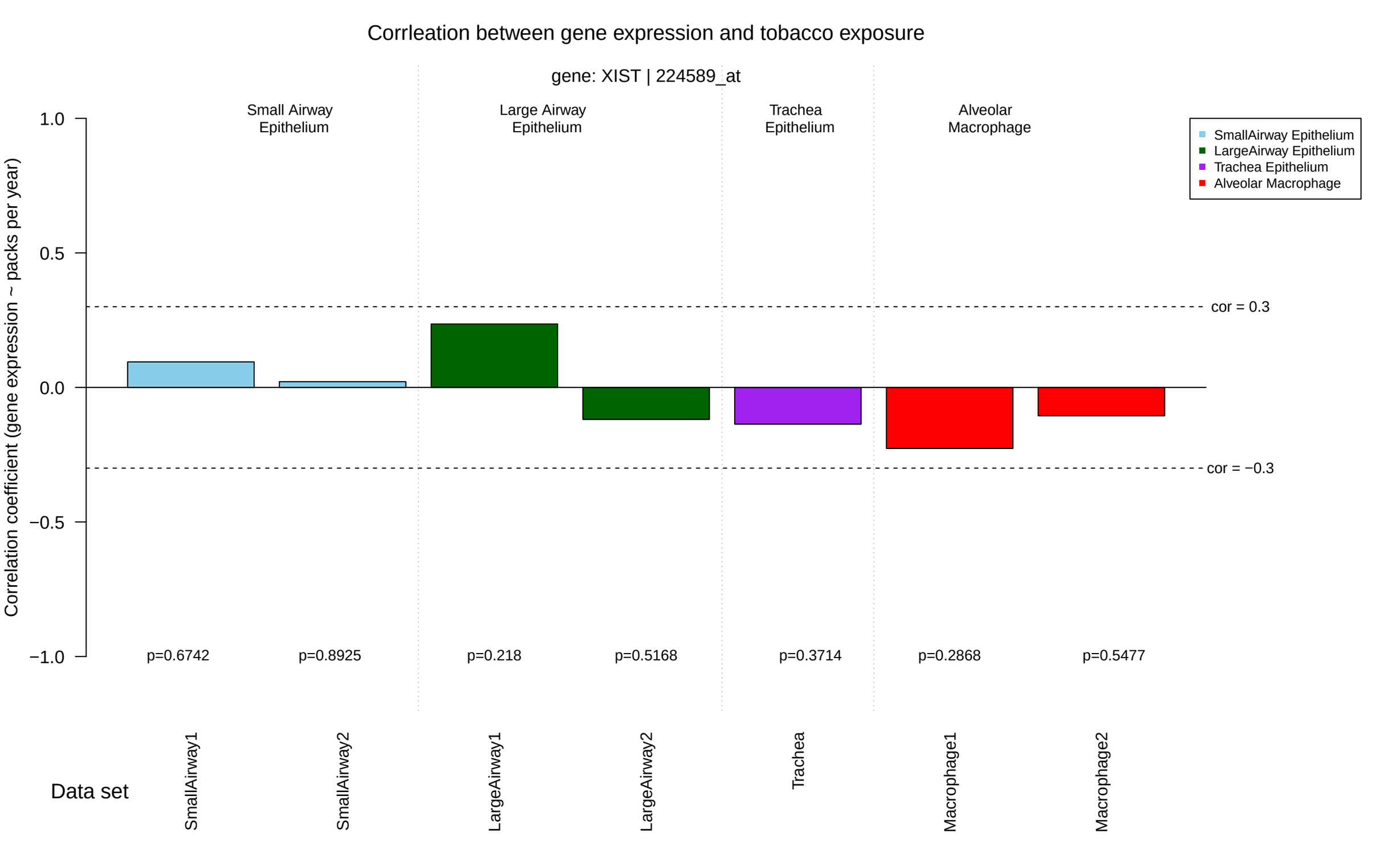

set: LargeAirway1 (p=0.0026)
XIST | 224589_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3979)
XIST | 224589_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2397)
XIST | 224589_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2404)
XIST | 224589_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9171)
XIST | 224589_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: XIST | 224589_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6742
p=0.8925
p=0.218
p=0.5168
p=0.3714
p=0.2868
p=0.5477
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2